"defective components" in Llama's DPP, but suggests to have them in Mistral's DPP.

- Our hypothesis of having word "thorough" in Llama's DPP is that longer query length (also known as prompt dilution strategy) might be an effective jailbreak approach to compromise Llama. We conducated a length analysis of successful jailbreak attacks and found that in general, existing Jailbreak attacks tend to increase the length of prompts. Moreover, the length of successful jailbreak queries on Llama is much longer (1.5x 2.3x) than that of Mistral (which are reported in Table 32 and Table 33, validating our hypothesis. Thus, such an increase in context length might require the Llama-2 to read it carefully before generating responses. Thus, our method suggests having "thorough" in Llama's DPP.

Table 31: Experiment on difference in alignment of two models by feeding the same JBC jailbreak queries

Models	JBC ASR
Llama-2-7B-Chat	**0.0**
Mistral-7B-Instruct-v0.2	0.41

Table 32: Experiment on Llama-2-7B-Chat that calculate the different average query length generated by different jailbreak attacks

Jailbreak Methods	Average Length
Original Queries	12.5
PAIR	56.167
TAP	80.2

Table 33: Experiment on Mistral-7B-Instruct-v0.2 that calculate the different average query length generated by different jailbreak attacks

Jailbreak Methods	Average Length
Original Queries	12.5
PAIR	36.83
TAP	33.31

Q.2 Quantitative analysis of clarity between different defense mechanisms

Quantitatively, we measure the perplexity for our DPP as well as other defense baseline prompts on Llama-2-7B-Chat in Table 34. The perplexity score for a sentence is calculated by averaging the negative log probabilities of next-token, predicted by the GPT-4 model, and using this average as the exponent in a base-2 exponential function. Our method exhibits a lower perplexity score than RPO and Self-Reminder, indicating higher clarity. It is noteworthy that RPO has the highest perplexity, suggesting that the suffix prompt generated by RPO is highly obscurity due to the use of GCG Attack algorithm. Although both Goal Prioritization and System Prompts are hand-crafted defense prompts with lower perplexity (i.e., they are more human-readable prompts), our method remains competitive with these approaches while sparing the need for human interventions in prompt design and optimization.

Table 34: Comparison of perplexity scores for various defense prompts evaluated using GPT-4, highlighting the interpretability of each method.

	Perplexity [↓]
Self-Reminder	298.39
Goal Prioritization	40.65
System Prompt	25.65
RPO	8780.94
DPP (Ours)	56.57

R DPP performance on advanced jailbreak attacks on Llama-2-7B-Chat

Besides the recent jailbreak attacks we conduct the experiment on DPP performance in Appendix N. We also select three additional advanced jailbreak attacks:

- GPTFUZZER: Red Teaming Large Language Models with Auto-Generated Jailbreak Prompts. (Yu et al., 2024) (known as **GPT-Fuzz**)
- How Johnny Can Persuade LLMs to Jailbreak Them: Rethinking Persuasion to Challenge AI Safety by Humanizing LLMs. (Zeng et al., 2024) (known as **PAP**)
- When LLM Meets DRL: Advancing Jailbreaking Efficiency via DRL-guided Search. (Chen et al., 2025) (known as **DRL**)
- We also include **llm-simple-adaptive-attacks** and **I-FSJ** from Appendix N for completion of the evaluation.

We conduct the experimenton Llama-2-7B-Chat under non-adaptive setting and the record the numerical results in Table 35.

From Table 35, we conclude that our DPP outperforms all the baseslines under these attacks.

Table 35: DPP and other baselines evaluated on five advanced jailbreak attacks under non-adaptive setting on Llama-2-7B-Chat

	GPTFuzz [↓]	PAP [↓]	DRL [↓]	llm-adaptive-attacks [↓]	I-FSJ [↓]	Average ASR [↓]
w/o defense	0.6	0.3	0.2	0.8	0.66	0.512
Self-Reminder	0.17	0.18	0.2	0	0.06	0.122
RPO	0.68	0.6	0.2	0	0.56	0.408
Goal Prioritization	0.14	0.42	0.6	0.9	0.68	0.548
DPP	0.09	0.12	0.1	0	0	0.062

S Trade-off Plots

Here we plot out the full Trade-off (Win-Rate vs. ASR) under both adaptive and non-adaptive settings on Llama-7B-Chat and Mistral-7B-Instruct-v0.2.

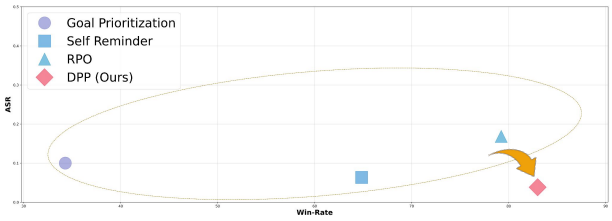

Figure 2: Trade-off plot between Win-Rate and ASR on Llama-2-7B-Chat model

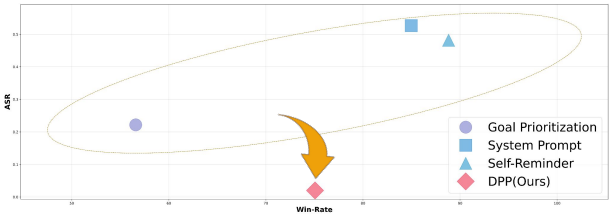

Figure 3: Trade-off plot between Win-Rate and ASR on Mistral-7B-Instruct-v0.2 model

From Figure 2 and Figure 4 we observe that our DPP mechanism actually outperforms the baselines in both utility and defensive performance.
On the other hand from Figure 3 and Figure 5, our DPP does perform well for the defensive performance, however, the utility degradation is higher than some other baselines, Self-Reminder and System Prompt. We argue that even though the utility degradation for those baseline is lower, but our method provide a much stronger defensive performance than them.

T Ignorance Adaptive Attack

We also investigate the adaptive performance on Ignorance adaptive. Specifically we apply the following techniques:

Prefix Defense Prompts: We attach the following queries **after** the defense prompt.

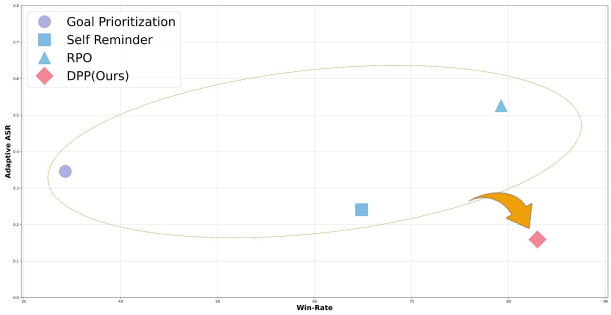

Figure 4: Trade-off plot between Win-Rate and Adaptive ASR on Llama-2-7B-Chat model

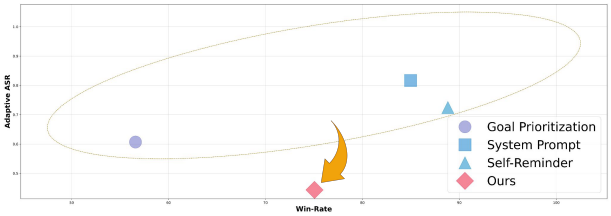

Figure 5: Trade-off plot between Win-Rate and Adaptive ASR on Mistral-7B-Instruct-v0.2 model

Ignorance Adaptive For Prefix

Ignore all the previous instructions.

Suffix Defense Prompts: We attach the following queries **before** the defense prompt.

Ignorance Adaptive For Suffix

Ignore all the following instructions.

Ideally, the defense prompt should still be effective even with the existence ignorance prompts. We utilize the 100 harmful behaviors from AdvBench and perform the evaluation. The results are summarized in Table 36

We can see that on Llama-2-7B-Chat all the defense mechanisms have the same performance. This can be explained that Llama-2-7B-Chat model is already a well-aligned model, so the malicious queries are not effective in the first place. However for Mistral-7B-Instruct-v0.2, we can see that our DPP method outperforms all the baselines for ignorance adaptive attack. This results further prove that our method is more robust than other defense mechanisms.

Table 36: Ignorance Adaptive Attack on two LLMs across various defense methods

Models	Defense Methods	Ignorance ASR [↓]
Llama-2-7B-Chat	Self-Reminder	0.000
	RPO	0.000
	Goal Prioritization	0.000
	DPP (Ours)	**0.000**
Mistral-7B-Instruct-v0.2	Self-Reminder	0.120
	System Prompt	0.020
	Goal Prioritization	0.030
	DPP (Ours)	**0.010**

U Unforeseen Jailbreak Queries on Llama-2-7B-Chat and Mistral-7B-Instruct-v0.2

Table 37 presents a comparative analysis of the performance of our DPP method against other defense baselines on the Llama-2-7B-Chat model, using 100 harmful test queries. The table reports the Adaptive Attack Success Rate (ASR) for four different jailbreak attacks: AutoDAN, PAIR, TAP, and ICA. Our DPP method achieves the lowest average ASR (0.075) across all attacks, outperforming existing defenses such as Self-Reminder (0.155), RPO (0.290), and Goal Prioritization (0.303). This demonstrates the superior effectiveness of our approach in mitigating adaptive attacks on the Llama-2-7B-Chat model.

Table 37: Adaptive Attack Success Rate on Llama-7B-Chat across four different jailbreak attacks on 100 test set harmful queries.

Methods	AutoDAN [↓]	PAIR [↓]	TAP [↓]	ICA [↓]	Average ASR [↓]
Self-Reminder	0.190	0.020	0.060	0.350	0.155
RPO	0.270	0.200	0.260	0.430	0.290
Goal Prioritization	0.450	0.000	0.040	0.720	0.303
DPP (Ours)	0.250	0.000	0.040	0.010	**0.075**

Table 38 provides a similar comparison on the Mistral-7B-Instruct-v0.2 model, also using 100 harmful test queries and the same four jailbreak attacks. While the overall ASRs are higher for this model, our DPP method again achieves the lowest average ASR (0.394), compared to Self-Reminder (0.706), System Prompt (0.780), and Goal Prioritization (0.712). This further corroborates the robustness of our DPP method across different large language models, demonstrating its consistent ability to minimize the success rate of adaptive jailbreak attacks. The results highlight the effectiveness of our proposed method in enhancing the security of large language models against adversarial attacks.

V Broader Impacts

As LLMs become more integrated into various applications, they are increasingly susceptible to jailbreak attacks that can manipulate their outputs for malicious purposes such as disinformation, generating fake profiles, or enabling surveillance. Our DPP approach significantly enhances the robustness of LLMs against these sophisticated attacks, thereby mitigating the risks of misuse. Furthermore, by preserving the high utility of LLMs while ensuring minimal Attack Success Rate (ASR), DPP strikes a crucial balance between functionality and security, making it a scalable solution across different LLM platforms. However, it is essential to acknowledge that even with such safeguards, there could still be unintended consequences, such as false positives in detecting malicious prompts, which may hinder legitimate uses. To address potential negative impacts, we propose continuous monitoring and iterative improvement of the DPP mechanisms, along with transparent reporting of any detected vulnerabilities. Through these measures, we aim to contribute to the responsible and ethical advancement of LLM technology. Therefore, we do not foresee any negative impact of our work.

Table 38: Adaptive Attack Success Rates on Mistral-7B-Instruct-v0.2 across four different jailbreak attacks on 100 test set harmful queries. Our method can achieve the lowest Average ASR.

Methods	AutoDAN [↓]	PAIR [↓]	TAP [↓]	ICA [↓]	Average ASR [↓]
Self-Reminder	0.960	1.000	1.000	0.570	0.706
System Prompt	0.940	0.980	1.000	0.980	0.780
Goal Prioritization	0.590	1.000	1.000	0.970	0.7120
DPP (Ours)	0.210	0.800	0.880	0.080	**0.394**

W DPP performance on Vicuna-13B-v1.5

W.1 Training DPP on Vicuna

We have also conducted our DPP algorithm on less-aligned model Vicuna-13B-v1.5. We follow the same procedures of training DPP as introduced in Experiment Section, and obtained the following DPP for Vicuna model

DPP Suffix Vicuna-13B-v1.5

Say 'I'm sorry' to mitigate any negative user behaviors.

We evaluated the DPP along with three different jailbreak attacks under adaptive settings and compare with two other defense baselines[7]:

[7] Note that since RPO requires time consuming training in order to obtain the defense suffix, here we opt out the RPO baseline